EXPERIMENT – 2

Logical, Bit-Manipulation and Shift Operations Using 8-bit and 16-bit Data Manipulation

Objective

To develop and execute 8085 assembly language programs for logical operations such as **AND, OR, XOR, NOT and two's complement**, along with bit masking, bit setting, clearing, testing, shifting and rotating operations.

Algorithm

1. Load the input data into the required registers.
2. Perform AND, OR and XOR operations.
3. Apply masks to obtain the lower and upper nibbles.
4. Perform one's and two's complement operations.
5. Set, clear and toggle required bits.
6. Perform rotate and shift operations.
7. Store the results in the specified memory locations.
8. Verify the results using bit-by-bit calculations.

Software and Hardware Required

- 8085 Simulator
- 8-bit Processor Trainer Kit
- Keyboard
- Power Supply

A) Logical and Bit Masking Operations

Code

```
LXI SP,47FFH
LXI H,4500H
MOV A,M
INX H
MOV B,M
ANA B
STA 4502H
```

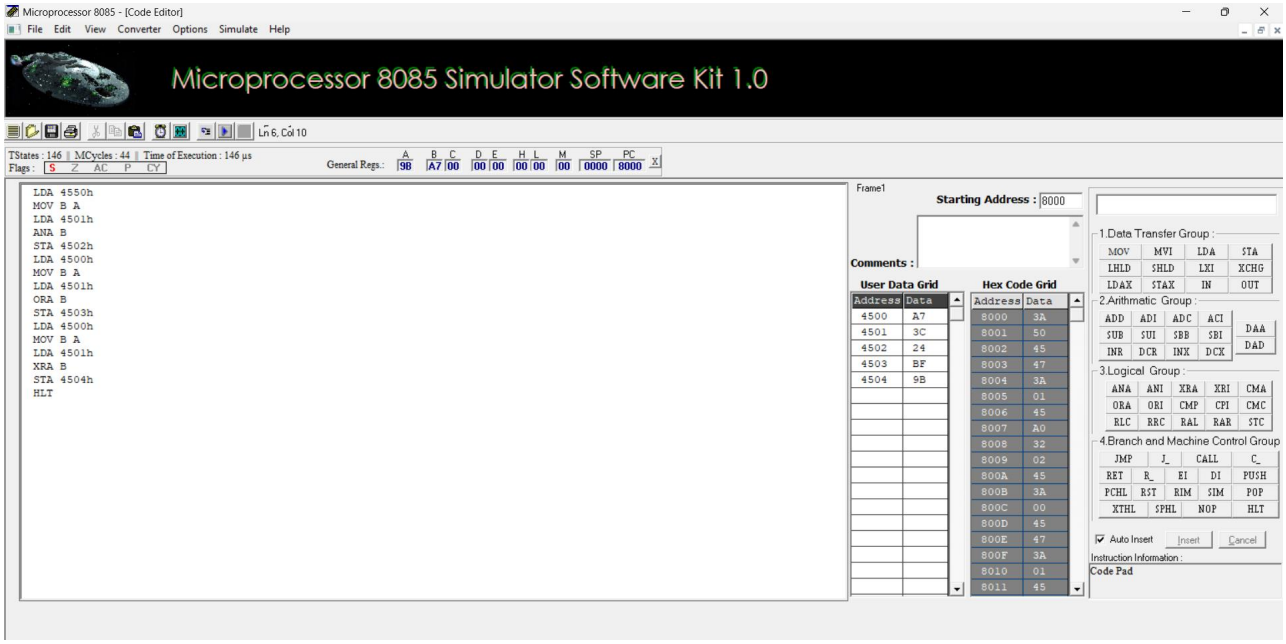
```
LDA 4500H
ORA B
STA 4503H
LDA 4500H
XRA B
STA 4504H
LDA 4500H
ANI 0FH
STA 4505H
LDA 4500H
ANI F0H
STA 4506H
LDA 4500H
CMA
STA 4507H
INR A
STA 4508H
HLT
```

The master record specifies Program A for AND, OR, XOR, nibble masking, one's complement and two's complement operations.

Input

Address	Data
4500H	A7H
4501H	3CH

Output



Output Address	Data
4502H	24H
4503H	BFH
4504H	9BH

Calculation:

- A7H AND 3CH = **24H**
- A7H OR 3CH = **BFH**
- A7H XOR 3CH = **9BH**
- A7H AND 0FH = **07H**
- A7H AND F0H = **A0H**
- NOT A7H = **58H**
- Two's complement = **59H**

B) Bit Setting, Clearing, Shift and Rotate Operations

Code

LDA 4510H

ORI 81H

STA 4511H

```
LDA 4510H
ANI F5H
STA 4512H
LDA 4510H
XRI 0FH
STA 4513H
LDA 4510H
ANI 40H
JZ CLEAR
MVI A,01H
JMP STORE
CLEAR:
MVI A,00H
STORE:
STA 4514H
LDA 4510H
RLC
STA 4515H
LDA 4510H
RRC
STA 4516H
LDA 4510H
ORA A
RAL
STA 4517H
LDA 4510H
ORA A
RAR
STA 4518H
```

LDA 4510H

RAR

STA 4519H

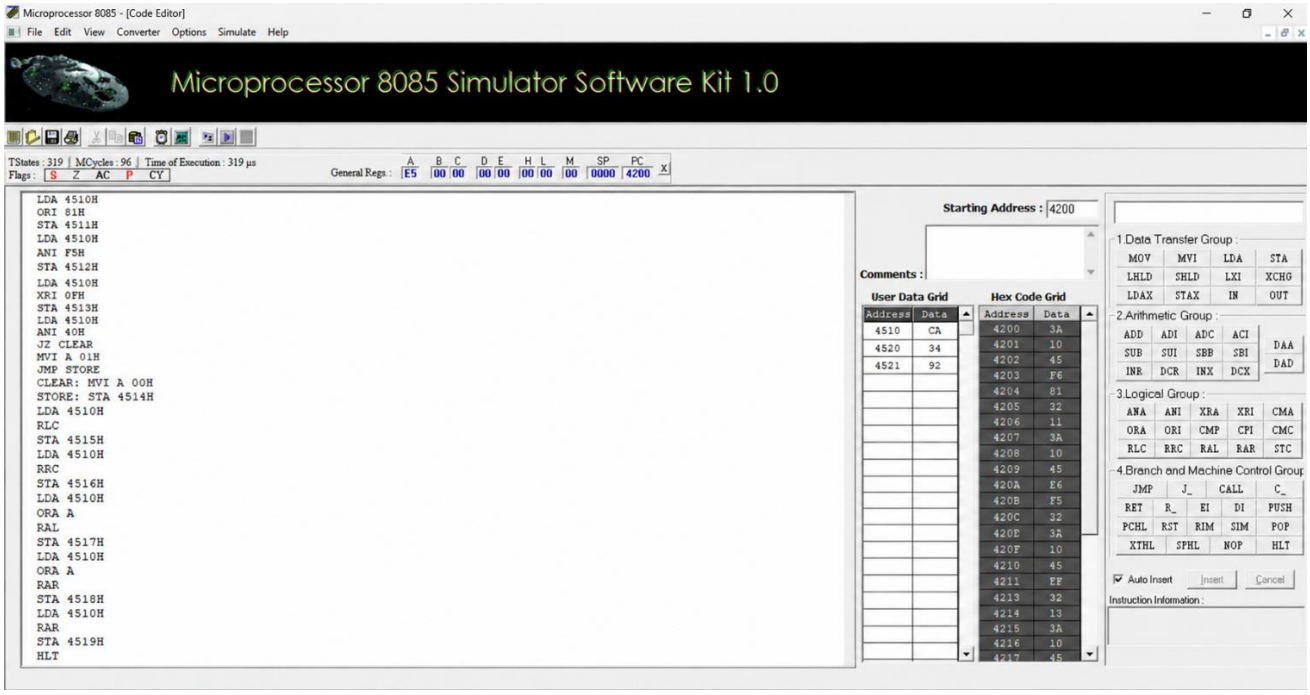
HLT

Program B performs bit setting, clearing, toggling, bit testing, rotate and shift operations.

Input

Address	**Data**
4510H	CAH
4520H–4521H	9234H

Output



RESULT

Thus, the **logical, bit-manipulation, masking, rotate and shift operations were successfully developed and verified using 8085 assembly language programming**. The obtained results were stored in the specified memory locations and verified bit by bit.

Student Reflection

I learned how to perform logical and bit-level operations using the 8085 microprocessor. I understood the use of AND, OR, XOR, masking, complement, bit setting, clearing, rotation and shifting instructions. This experiment helped me understand how individual bits can be manipulated and verified using assembly language.